Index notation

1 State the base for the expression 9^5.

2 State the power for the expression 2^7.

3 Write the following powers of 9 in index form:

a 9×9

b $9 \times 9 \times 9$

c $9 \times 9 \times 9 \times 9$

d $9 \times 9 \times 9 \times 9 \times 9$

4 Write the following expressions using index notation:

a $5 \times 5 \times 5$

b $4 \times 4 \times 4 \times 4 \times 4$

c $(-9) \times (-9) \times (-9) \times (-9)$

d 7×7

e $12 \times 12 \times 9 \times 9 \times 9$

f $(-13) \times (-13) \times (-13) \times (-13) \times 5 \times 5$

g $11 \times 7 \times 11 \times 7 \times 7$

h $3 \times (-4) \times 3 \times 3 \times (-4)$

i $(-6) \times (-2) \times (-6) \times (-6) \times (-2)$

j $12 \times 12 \times 12 \times 15 \times 15$

k $2 \times 4 \times 2 \times 4$

l $24 \times 24 \times 24 \times 12$

5 Write the following statements using index notation:

a Two squared

b Three cubed

6 Evaluate the following:

a 2^6

b $(-3)^3$

c $3^3 \times 3^2$

d $2^3 \times 4^2$

e $\dfrac{3^5}{3^3}$

f $\dfrac{(-6)^9}{6^7}$

g $2^2 + 4^3$

h $7^2 + 3^4$

i $12^2 - 2^2$

j $5^3 - 3^3$

k $3^3 - 2^2 \times 4^2$

l $(-6)^2 - 3^8 \div 3^6$

Expanded notation

7 Write the following expressions in expanded form:

a 9^2

b 9^3

c 9^4

d 9^5

8 Complete the table below:

Words	Expanded notation	Index notation
Six to the power of four		
Eight cubed		
Eleven to the power of three		
Twenty three to the power of five		
Seven squared		
Fifteen to the power of four		

9 Write the following numbers in expanded form:

a 5^2 b $7^5 \times 6^4$ c $4^6 \times 9^2$ d $77^5 \times 13^2$

10 Consider the expression 5^3.

a Write the expression in expanded form.

b Using your answer to part (a), find the value of 5^3.